

SHAOCHENG LUO

<https://shaochengluo.github.io/>

Room 435 Wilkinson Building, Duke University, Durham, North Carolina 27705, USA
(408) 306 7312 ◊ shaocheng.luo@duke.edu

RESEARCH INTERESTS

My research focuses on advancing **Trustworthy Autonomous Systems** and **Multi-Robot Systems** by enhancing the performance of low-cost, resource-constrained devices and employing collaborative perception for enhanced system integrity. These advancements are particularly critical in applications such as autonomous driving and drone operations, where trustworthiness—built on *robustness*, *resilience*, and *security*—is essential for ensuring dependable and safe autonomous behavior in dynamic, unpredictable, and contested environments.

EDUCATION

Postdoctoral Associate, NSF AI Institute for Edge Computing (Athena), Department of Electrical and Computer Engineering Aug 2022 - present

Duke University, Durham, North Carolina, USA

Advisor: Dr. Miroslav Pajic

Postdoctoral Fellow, Alberta Autonomous Systems Initiative (ASI), Department of Computing Science Aug 2020 - Mar 2022

University of Alberta, Edmonton, Alberta, Canada

Advisor: Dr. Hong Zhang

Ph.D., Computer and Information Technology Aug 2014 - May 2020

Purdue University, West Lafayette, Indiana, USA

Dissertation: Multi-robot System in Coverage Control: Deployment, Coverage, and Rendezvous

Advisor: Dr. Byung-Cheol Min

M.S., Mechatronic Engineering Sept 2009 - Mar 2012

Zhejiang University, China

Thesis: Design of Multi-axis Motion Controller Based on Fieldbus Communications

Advisor: Dr. Shiqiang Zhu

B.S., Mechanical Engineering Aug 2005 - Jul 2009

Harbin Institute of Technology, China

PUBLICATIONS

Journal Papers:

- [1] **Shaocheng Luo**, Jonghoek Kim, and Byung-Cheol Min, “Asymptotic Boundary Shrink Control with Multi-robot Systems”, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*. Vol. 52 (1), 591-605, 2020.
- [2] Ramviyas Parasuraman, Jonghoek Kim, **Shaocheng Luo**, and Byung-Cheol Min, “Multi-Point Rendezvous in Multi-Robot Systems”, *IEEE Transactions on Cybernetics*, Vol. 50 (1), pp. 310-323, 2020.
- [3] **Shaocheng Luo**, Jonghoek Kim, Ramviyas Parasuraman, Jun Han Bae, Eric T Matson, and Byung-Cheol Min, “Multi-robot Rendezvous Based on Bearing-aided Hierarchical Tracking of Network Topology”, *Ad Hoc Networks*, Vol 86, pp. 131-143, 2019.
- [4] Xinglai Jin, Shiqiang Zhu, Wenxiang Wu, and **Shaocheng Luo**, “A Novel Robotic Motion Control Strategy Based on Improved Fuzzy PID and Feedforward Compensation”, *Applied Mechanics and Materials*, Vol. 365-366, pp. 821-826, 2013.

- [5] **Shaocheng Luo**, Shiqiang Zhu, and Huifang Wang, “Adaptive PID saturate output feedback control of robot manipulators”, *Transducer and Microsystem Technologies (Chinese)*, Vol. 31 (3), pp. 66-70, 2012.

Conference Papers:

- [1] Haocheng Meng, **Shaocheng Luo***, Zhenyuan Liang, Qing Huang, Amir Khazraei, and Miroslav Pajic. “MARS: Defending Unmanned Aerial Vehicles From Attacks on Inertial Sensors with Model-based Anomaly Detection and Recovery.” *2025 IEEE Symposium on Security and Privacy (S&P)* (Under review).
- [2] David Hunt, **Shaocheng Luo***, Spencer Hallyburton, Yi Li, Ray Hoggard, Aarav Prakash, Ryan Bauroth, Tingjun Chen, and Miroslav Pajic. “RadNav: mmWave Radar-based Localization and Navigation for Unmanned Ground Vehicles in Complex Indoor Environments.” *2025 ICRA* (Under review).
- [3] Xiao Zhang, David Hunt, **Shaocheng Luo**, Miroslav Pajic, and Tingjun Chen. “mmUnion: High Resolution 360-degree Point Cloud Generation via Multi-Radar Fusion on Unmanned Ground Vehicles.” *2025 ICRA* (Under review).
- [4] David Hunt, **Shaocheng Luo***, Amir Khazraei, Xiao Zhang, Spencer Hallyburton, Tingjun Chen, and Miroslav Pajic. “RadCloud: Real-Time High-Resolution Point Cloud Generation Using Low-Cost Radars for Aerial and Ground Vehicles.” *2024 ICRA*, Yokohama, Japan, May 13-17, 2024.
- [5] Hao-Lun Hsu, Haocheng Meng, **Shaocheng Luo**, Juncheng Dong, Vahid Tarokh, and Miroslav Pajic. “REFORMA: Robust REinFORceMent Learning via Adaptive Adversary for Drones Flying under Disturbances.” *2024 ICRA*, Yokohama, Japan, May 13-17, 2024.
- [6] **Shaocheng Luo**, Yogang Singh, Hanyao Yang, Jun Han Bae, J. Eric Dietz, Xiumin Diao, and Byung-Cheol Min, “Image Processing and Model-Based Spill Coverage Path Planning for Unmanned Surface Vehicles”, *2019 MTS/IEEE OCEANS*, Seattle, WA, USA, October 27-31, 2019.
- [7] Jun Han Bae, **Shaocheng Luo**, Yogang Singh, Bumjoo Lee, Richard M. Voyles, Mauricio Postigo, Edgar Gonzales, Lizbeth Paredes Aguilar, and Byung-Cheol Min, “Development of an Unmanned Surface Vehicle for Remote Sediment Sampling with a *Van Veen* Grab Sampler”, *2019 MTS/IEEE OCEANS*, Seattle, WA, USA, October 27-31, 2019.
- [8] Manoj Penmetcha, **Shaocheng Luo**, Arabinda Samantaray, J. Eric Dietz, Baijian Yang, and Byung-Cheol Min, “Computer Vision-based Algae Removal Planner for Multi-robot Teams”, *2019 IEEE International Conference on Systems, Man and Cybernetics (SMC)*, Bari, Italy, October 6-9, 2019.
- [9] **Shaocheng Luo**, Jun Han Bae, and Byung-Cheol Min, “Pivot-based Collective Coverage Control with a Multi-robot Team”, *2018 IEEE International Conference on Robotics and Biomimetics (IEEE ROBIO 2018)*, Kuala Lumpur, Malaysia, December 12-15, 2018.
- [10] Walter D. Leon-Salas, Thomas Fischer, Xiaozhe Fan, Golsa Moayeri, and **Shaocheng Luo**, “A 64x64 Image Energy Harvesting Configurable Image Sensor”, *IEEE International Symposium on Circuit and Systems (ISCAS 2016)*, Montreal, Canada, May 22-25, 2016.
- [11] **Shaocheng Luo** and Shiqiang Zhu, “Open Architecture Multi-Axis Motion Control System for Industrial Robot Based on CAN Bus”, *Automatic Control and Artificial Intelligence (ACAI 2012)*, Xiamen, China, March 24-26, 2012.

Reports and Presentations:

- [1] **Shaocheng Luo**, “Multi-robot System Control and Collective Perception for Environmental Monitoring and Operations”, *Research Presentation*, Oregon State University, Virtual, February 23, 2022.
- [2] Siqi Yan, Xiaolong Wang, **Shaocheng Luo**, and Hong Zhang, “Collective Driving with Shared Perception to Improve Driving Safety”, *Autonomous Systems Initiative (ASI) Real-World Automated Transport Workshop*, University of Alberta, Edmonton, AB, Canada, October 2, 2020.

- [3] **Shaocheng Luo**, “Coverage Control Using Multi-robot Systems and Its Applications in Environmental Monitoring and Operations”, *Electrical and Computer Engineering Seminar*, University of Louisville, Louisville, KY, USA, March 6, 2020.
- [4] **Shaocheng Luo**, “Multi-robot Systems and Their Applications for Greater Human Well-being”, *Global Scientist Interdisciplinary Forum at Southern University of Science and Technology (SUSTech)*, Shenzhen, China, January 4, 2020.
- [5] **Shaocheng Luo**, “Advances of Multi-robot Systems in Environmental Monitoring and Operations: Three Basic Collective Behaviors”, *Shanghai University 2019 International Forum for Young Scholars on Intelligent Vehicle Science & Engineering*, Shanghai, China, November 12, 2019.
- [6] **Shaocheng Luo**, Yogang Singh, Hanyao Yang, Junhan Bae, J. Eric Dietz, Xiumin Diao and Byung-Cheol Min, “Coverage Path Planning for Efficient Spill Cleaning in Maritime Environment”, *6th Annual Environmental Community Mixer, Purdue Discovery Park Center for the Environment (C4E)*, Purdue University, West Lafayette, IN, USA, October 17, 2019.
- [7] Yogang Singh, Junhan Bae, Shyam Sundar Kannan, **Shaocheng Luo**, Wonse Jo, Yuta Haoshi, Jose Garcia, Brittany Newell, Mauricio Postigo, Sara McMillan, Richard Voyles, Lisabeth Leonor Pardes Aguilar, Godofredo Pena, Edgar Gonzales Zenteno and Byung-Cheol Min, “Design and Development of Unmanned Robotic Water Quality Monitoring and Sediment Sampling Systems”, *6th Annual Environmental Community Mixer, Purdue Discovery Park Center for the Environment (C4E)*, Purdue University, West Lafayette, IN, USA, October 17, 2019.
- [8] **Shaocheng Luo**, “Multi-robot Systems with Applications in Water Environmental Monitoring and Operations”, *Research Presentation in Senseable City Lab at MIT*, Cambridge, MA, USA, October 1, 2019.
- [9] Junhan Bae, Yogang Singh, **Shaocheng Luo**, Wonse Jo, Yuta Haoshi, Jose Garcia, Brittany Newell, Mauricio Postigo, Sara McMillan, Richard Voyles, Lisabeth Leonor Pardes Aguilar, Godofredo Pena, Edgar Gonzales Zenteno and Byung-Cheol Min, “Robotic Water Quality Monitoring and Sediment Sampling: A Pilot Study”, *2019 NEXUS Workshop Poster Presentation*, Purdue University, West Lafayette, IN, USA, July 25, 2019.
- [10] **Shaocheng Luo**, Wonse Jo, Jun Han Bae, and Byung-Cheol Min, “A Systematic Solution for Autonomous Aquatic Spill Removal with a Multi-robot Team”, *Fall 2018 Realizing the Digital Enterprise (RDE) Graduate Research Poster Session*, West Lafayette, IN, USA, November 26, 2018.
- [11] **Shaocheng Luo** and Byung-Cheol Min, “Algae Harvesting with a Multi-robot Team”, *Purdue Dawn or Doom ’18 Emerging Technology Conference*, West Lafayette, IN, USA, November 5-6, 2018.
- [12] **Shaocheng Luo**, Jun Han Bae, and Byung-Cheol Min, “Oil Spill Cleanup with Multi-robot Systems”, *5th Annual Environmental Community Mixer, Purdue Discovery Park Center for the Environment (C4E)*, West Lafayette, IN, USA, October 12, 2018.
- [13] **Shaocheng Luo**, Ramviyas Parasuraman, Jun Han Bae, and Byung-Cheol Min, “Environmental Operations with Multi-robot Systems”, *2018 Midwest Robotics Workshop (MWRW 2018)*, Chicago, IL, USA, June 14-15, 2018.
- [14] **Shaocheng Luo**, Tamzidul Mina, Ramviyas Parasuraman, Byung-Cheol Min, “Distributed Algorithms for Coordination of Multi-Robot Systems”, *Polytechnic Faculty Convocation 14th Annual Poster Session*, West Lafayette, IN, USA, March 31, 2018.
- [15] **Shaocheng Luo**, Ramviyas Parasuraman, Jun Han Bae, Jonghoek Kim, and Byung-Cheol Min, “Robust Multi-Robot Rendezvous Control in Cluttered and GPS-Denied Environments”, *Cocktails & Collaboration with Northrop Grumman Poster Session*, West Lafayette, IN, USA, September 13, 2017.
- [16] **Shaocheng Luo**, Ramviyas Parasuraman, Jun Han Bae, Sangjun Lee, Jonghoek Kim, and Byung-Cheol Min, “Multi-robot Rendezvous Control and Optimization”, *2017 Midwest Robotics Workshop (MWRW)*

2017), Chicago, IL, USA, May 18-19, 2017.

- [17] **Shaocheng Luo**, Ramviyas Parasuraman, Jun Han Bae, Jonghoek Kim, and Byung-Cheol Min, "Time-Constraint Multi-Robot Rendezvous Control and Optimization", *2017 NSF I/UCRC: Center for Robots and Sensors for the Human Well-being (RoSe-HUB) IAB Meeting*, Minneapolis, MN, USA, April 26-28, 2017.
- [18] **Shaocheng Luo**, Jun Han Bae, Ramviyas Parasuraman, Jonghoek Kim, and Byung-Cheol Min, "Bounded Distributed Rendezvous Control while Preserving Connectedness in Cluttered Environments", *2016 NSF I/UCRC: Center for Robots and Sensors for the Human Well-being (RoSe-HUB) IAB Meeting*, Charlotte, NC, USA, November 17-18, 2016.
- [19] Jun Han Bae, **Shaocheng Luo**, Jee Hwan Park, Dong Hun Lee, and Byung-Cheol Min, "Water Quality Monitoring System Based on Robotics and Cyber-Physical Systems", *2016 NSF I/UCRC: Center for Robots and Sensors for the Human Well-being (RoSe-HUB) IAB Meeting*, Charlotte, NC, USA, November 17-18, 2016.

SCHOLARSHIPS AND AWARDS

- Purdue Polytechnic Dean's Graduate Student Travel Grant, *Oct 2019*. Amount: \$300
- Purdue Computer and Information Technology Graduate Student Travel Grant, *Oct 2019*. Amount: \$600
- International Travel Support Funds from Purdue Polytechnic Office of Globalization, *Oct 2019*. Amount: \$600
- Purdue Graduate Student Government Travel Grant, *Dec 2018*. Amount: \$500
- Purdue Polytechnic Dean's Graduate Student Travel Grant, *Nov 2018*. Amount: \$600
- Purdue Computer and Information Technology Graduate Student Travel Grant, *Oct 2018*. Amount: \$600
- International Travel Support Funds from Purdue Polytechnic Office of Globalization, *Oct 2018*. Amount: \$750
- 2018 Polytechnic Institute Summer Research Scholarship. Amount: \$3,333 (Provided by the Graduate School during *May 1 - Aug 15, 2018*)
- 2016 Xinjiang Government Award for Outstanding Self Finance Students Abroad (Awarded by Education Office, Consulate-General of the P. R. China in Chicago), *May 2017*
- SMART Lab 2016 Yearly Most Valuable Person Award from lab director Dr. Byung-Cheol Min, *Jan 2017*
- Kaggle Competition: Shelter Animal Outcomes Prediction (Top 20%), *May 2016 - June 2016*
- Full Graduate Student Scholarship, Zhejiang University, *Sep 2009 - Mar 2012*
- Kindness Ambassador, Hangzhou Rehabilitation Center for Hearing-impaired Children (HRCHC), *2011*
- Merit Student of Zhejiang University, *2010*
- Provincial Excellent Graduate, Harbin Institute of Technology, *2009*
- Merit Student of Harbin Institute of Technology, *2007 and 2008*
- First Class the People's Scholarship, Harbin Institute of Technology, *2007*

TEACHING

CNIT 161: Introduction to Programming and Data Management for Smart Manufacturing (Sole Instructor)

Spring 2020

Dept of Computer and Information Technology, Purdue, West Lafayette

- Python and SQL languages are introduced as a combination in this course, for the purpose of database operations.
- Methodologies such as data analysis and business status investigation are introduced and practiced in the scenario of manufacturing.

CNIT 105: Introduction to C Programming Language (Lab Instructor)

Fall 2019, Spring 2019, Fall 2018, Spring 2018, Fall 2017

Dept of Computer and Information Technology, Purdue, West Lafayette, Supervisor: Prof. Guity Ravai

- Average Course Evaluation: 4.6/5.0 (Department Average: 3.0/5.0); Sessions taught per semester: 4; Average number of students per session: 25.
- Designed inlabs, weekly assignments, and exams.
- Was the first one who introduced debugging exercise to students in lab sessions, and got tremendous positive feedback.
- Instructed 6 undergrad lab helpers, because we sometimes recruit a few lab helpers from previous students to provide more immediate response and assistance for new students.

STEM Academic Boot Camp: Introduction to Computer Programming (Sole Instructor)

Summer 2019

Polytechnic Institute, Purdue, West Lafayette, Coordinator: Ms. Antonia Munguia

- This is the course created in Purdue to support the education of first-year college students from underrepresented socioeconomic regions and communities and women in the STEM field. This course met every working day for 1.5 hours and lasted for five weeks. I designed the lectures, labs, and exams.

Introduction to Microprocessors (Sole Instructor)

Fall 2016

SMART Lab, Purdue, West Lafayette, Supervisor: Dr. Byung-Cheol Min

- This is a eight-week long seminar based course for SMART Lab members, including junior PhDs, Masters, and Undergrads, to help them gain knowledge in microprocessors and their applications in Robotics.
- I designed the course and labs, with a specific topic every week on structure of microprocessor, hardware utilities, communication mechanism and protocols, sensor reading, motion control, etc.

GRANT WRITING EXPERIENCES

External Research Grants

• Awarded

- DURIP 2023: Validation Testbed for Trust-Aware Autonomy in Contested Environments, Sponsor: Office of Naval Research, PI: Miroslav Pajic, Amount: **\$450,000**, 2023 (Assisted in writing)
- CAREER: Adaptive Human Multi-robot Systems, Sponsor: National Science Foundation, PI: Byung-Cheol Min, Amount: **\$500,000**, 2019 (Assisted in writing)
- UNSA NEXUS: Robotic Water Quality Monitoring and Distribution Systems: A Pilot Study, Sponsor: Universidad Nacional de San Agustin, PI: Byung-Cheol Min, Amount: **\$365,439**, 2017 (Assisted in writing)

• Not Awarded

- CPS: Medium: Multi-Robot Cyber-Physical Systems for Water Quality Monitoring, Sponsor: National Science Foundation, PI: Byung-Cheol Min, Amount: **\$999,638**, 2017 (Assisted in writing)

- NRI: FND: ARMOUR: Augmented Reality Modus Operandi for Multi-User, Multi-Robot Teams, Sponsor: National Science Foundation, PI: Byung-Cheol Min, Amount: **\$686,049**, 2017 (Assisted in writing)

Internal Research Grants

- **Awarded**

- Purdue Research Foundation (PRF) Summer Faculty Research Grant for 2017: CRAWLER: A Cyber-Physical System for Read-time Autonomous Water and Lakebed Environment Reporting, Sponsor: Purdue University, PI: Byung-Cheol Min, Amount: **\$8,000**, 2017 (Assisted in writing)

MENTORING EXPERIENCE

Graduate Student Mentoring

- **David Hunt** (PhD student with ECE at Duke University, Durham, NC) *Aug 2022 - present*
- **Haocheng Meng** (PhD student with ECE at Duke University, Durham, NC) *Aug 2022 - present*
- **Zhenyuan Liang** (Master's student with ECE at Duke University, Durham, NC) *Apr 2024 - present*
- **Qing Huang** (Master's student with ECE at Duke University, Durham, NC) *Apr 2024 - present*
- **Yi Li** (Master's student with ECE at Duke University, Durham, NC) *May 2024 - present*
- **Siqi Yan** (Master's student with CS at the University of Alberta, Canada) *Aug 2020 - Mar 2022*

Undergrad Student Mentoring

- **Hanyao Yang** (Now a Senior Undergrad student in Huaqiao University, China) *July 2019 - Sept 2019*
- **Sangjun Eom** (Now a Master student in CRL, Purdue, West Lafayette) *Aug 2015 - Jan 2016*
- **Taa Soratana** (Now a PhD student in the University of Michigan (Ann Arbor)) *Aug 2015 - Jan 2016*

ACADEMIC SERVICES

Journal Referee

- IEEE Transactions on Systems, Man and Cybernetics: Systems
- IEEE/ASME Transactions on Mechatronics
- IEEE Robotics and Automation Letters (RA-L)
- Journal of Intelligent and Robotic Systems
- International Journal of Advanced Robotic Systems
- International Journal of Adaptive Control and Signal Processing (ACSP)
- Computers & Security (COSE)
- Optimization and Engineering (OPTE)
- International Journal of Adaptive Control and Signal Processing
- International Journal of Advanced Manufacturing Technology

Conference Referee

- IEEE International Conference on Robotics and Automation (ICRA), 2019, 2020, 2021, 2024, & 2025.
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024.

- IEEE International Conference on Robotic Computing (IRC), 2018, 2019 & 2020.
- International Conference on Practical Applications of Agents and Multi-Agent Systems (PAAMS), 2020.
- International Conference on Social Robotics (ICSR), 2016.

PROFESSIONAL ACTIVITIES

- **Program Committee Member (Posters and Demos)**, ACM/IEEE International Conference on Cyber-Physical Systems, Irvine, California, USA, 2025.
- **Guest Editor**, Frontiers in Control Engineering: Networked Control Section, 2022.
- **Program Committee Member**, IEEE International Conference on Robotics and Biomimetics (RO-BIO), Dali, China, 2019.
- **Assisted in writing**, A Roadmap for US Robotics: From Internet to Robotics (2016 Edition), §6.2.2. *Environmental Monitoring*, 2016.

PROFESSIONAL MEMBERSHIPS

- | | |
|--|-----------------------|
| Institute for Electrical and Electronics Engineers (IEEE), Member | 2014 - present |
| <ul style="list-style-type: none"> • Technical Committee on Multi-Robot Systems (2019 - present) • IEEE Robotics and Automation Society (2014 - present) | |